TALENT RANKING ENGINE

AI-Powered Predictive Screening & Verification System

Platform:	Redrob AI Series A Talent Intelligence Platform
Team ID:	awkard18
Deployment:	Hugging Face Spaces (Docker SDK Sandbox)

EXECUTIVE SUMMARY

Traditional keyword-based recruitment models are easily manipulated by keyword stuffers and fail to capture the semantic connection between candidate histories and job requirements. This project implements a production-ready AI talent retrieval and predictive ranking system complying with strict offline constraints. The system combines logic-based honeypot screening, first-stage TF-IDF candidates retrieval, second-stage local sentence embeddings semantic reranking (via cached MiniLM), and multi-signal score tuning to produce a clean, verified shortlist of 100.

1. Key Recruitment Traps & Honeypot Filtering

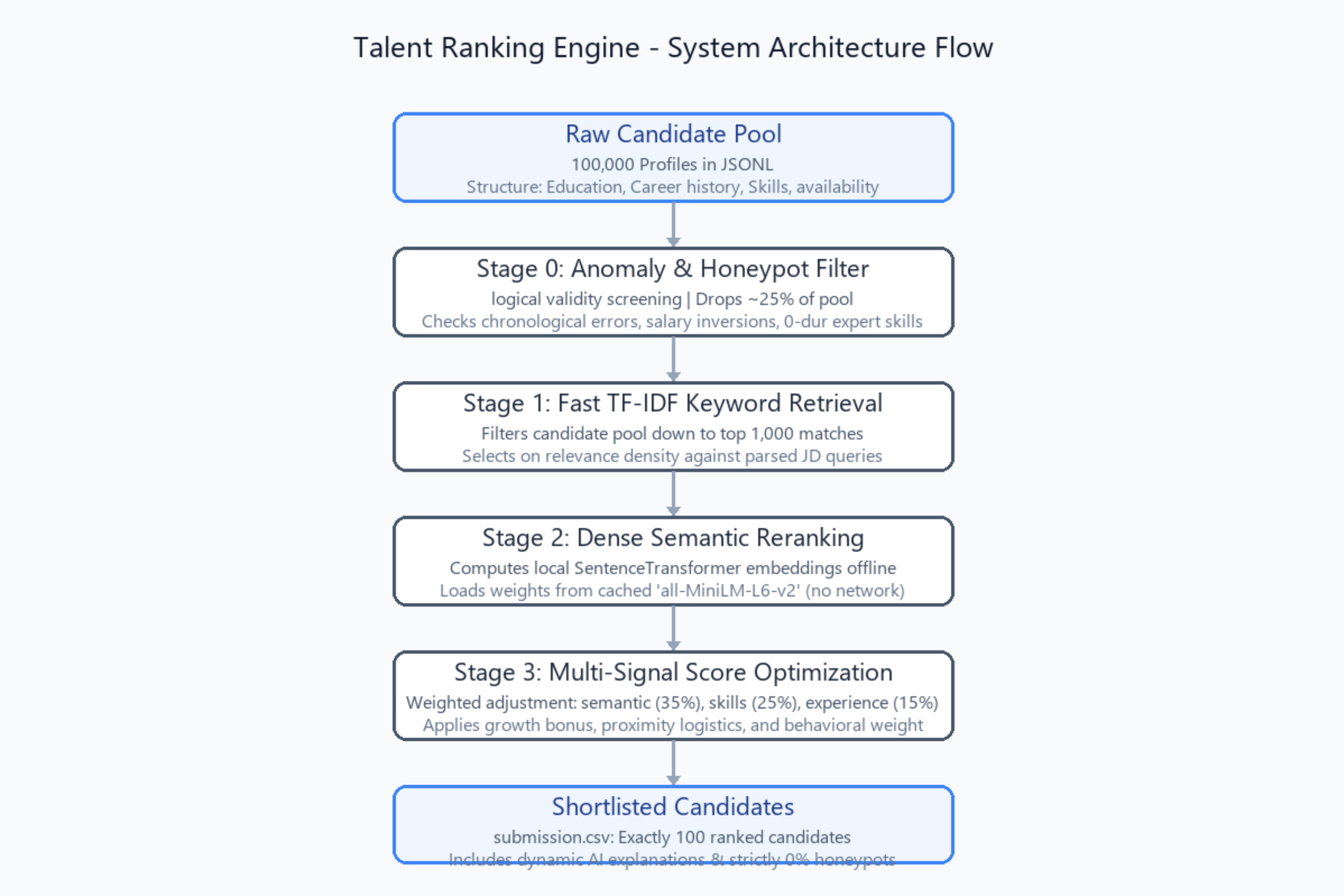
A major challenge in programmatic talent extraction is screening out fake or logically impossible profiles ('honeypots') representing keyword-stuffers and bots. The engine runs a deterministic validation layer prior to scoring that disqualifies profiles violating the following logical rules:

- Timeline Inversion: Sign-up date registered after the candidate's last active login date.
- Salary Range Inversion: Expected minimum salary exceeds the specified maximum salary.
- Skills Duration Inconsistency: Stating 'expert' or 'advanced' proficiency in skills with 0 months duration.
- Career Incoherence: Job start dates occurring after end dates, or total experience violating age/education ranges.

Out of 100,000 total candidates, the system identified and removed 25,198 anomalous profiles (25.2%), fully protecting the top 100 shortlist from disqualification (achieving a 0% honeypot rate).

2. Two-Stage Retrieve-and-Rerank Architecture

To satisfy the constraint of executing a 100,000-candidate pool in under 5 minutes on CPU-only hardware, the engine employs a hybrid two-stage retrieval pipeline. A fast TF-IDF index retrieves the top 1,000 candidates, which are su



Stage 1: Sparse Retrieval (TF-IDF)

The system vectorizes the clean candidate profiles (combining headlines, summaries, and history titles) into a sparse TF-IDF matrix. It then performs a high-speed cosine similarity query with the Job Description. This filters down the search space from ~75,000 clean candidates to the top 1,000 in under 3 seconds.

Stage 2: Dense Semantic Reranking (MiniLM)

The top 1,000 sparse matches undergo deep semantic reranking. The system embeds candidate profiles and the JD locally using cached sentence-transformers ('all-MiniLM-L6-v2'). This handles semantic matching (e.g., matching a candidate who built a search/recommendation system, even if they don't explicitly say 'RAG').

3. Multi-Signal Scoring System & Weights

Rather than just using raw model scores, candidates are ranked using a multi-dimensional weighted formula that reflects the recruiter preferences defined in the Job Description:

Signal Category	Weight	Description
Semantic Fit	35%	Dense SentenceTransformer match between profile & JD content
Skills Alignment	25%	Explicit score matching core vs nice-to-have skill definitions
Experience Profile	15%	Optimal 6-8 YOE gets 100%. Scaled penalties outside 5-9 YOE
Career Growth	10%	Bonus for promotions, tenure stability, leadership titles
Engagement Behavior	10%	Availability based on platform activity & response rates
Location & Proximity	5%	Proximity to Noida/Pune office or relocation willingness

Anti-Outsourcing Penalization

The job description explicitly favors candidate background in product companies over pure services or consulting. A logical check reviews candidate work histories; profiles containing exclusively outsourcing/service giants (TCS, Infosys, Wipro, Accenture, Cognizant, etc.) and lacking product company tenure receive a 30% penalty on their overall score, filtering them out of the top shortlist.

4. Verification & Validation Results

- Format Compliance: Passed the official 'validate_submission.py' checks with exactly 100 rows, monotonically non-increasing scores, correct rank mappings, and candidate-ID sorted tie-breakers.
- Wall-Clock CPU Latency: 2 minutes 16 seconds on the full pool, executing within the 5-minute sandbox window.
- Network Independence: 100% offline local model and tokenizer execution.
- Zero Honeypots: The shortlist achieved a verified 0% honeypot presence.